Pipelining and Asynchronous Input Handling in FPGA

Anirudh Chaturvedi (SR 25850) DESE, IISc Bangalore

Target Device: Artix-7 XC7A35T

1 Objective

To study pipelining and handling asynchronous inputs using synchronization and debouncing.

2 Pipelined Radix-4 Booth Multiplier

2.1 RTL Design

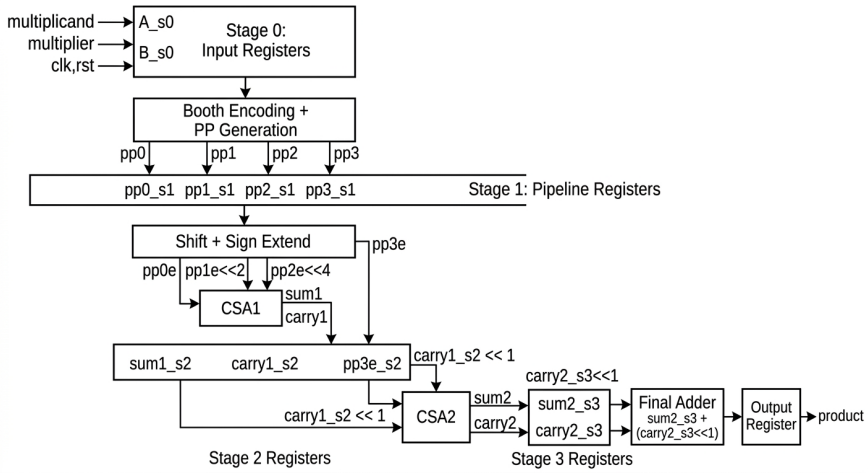


Figure 1: Intuition based Block Diagram

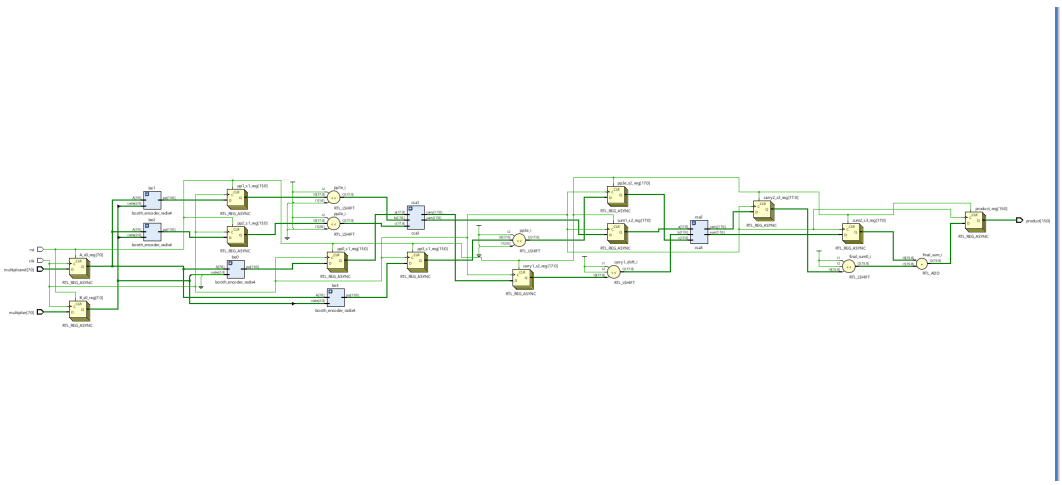


Figure 2: RTL Schematic

2.2 Simulation

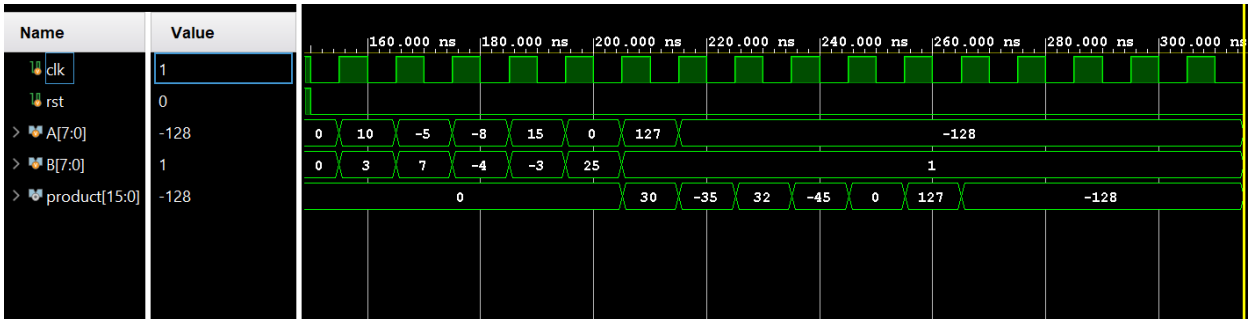


Figure 3: Behavioral Simulation

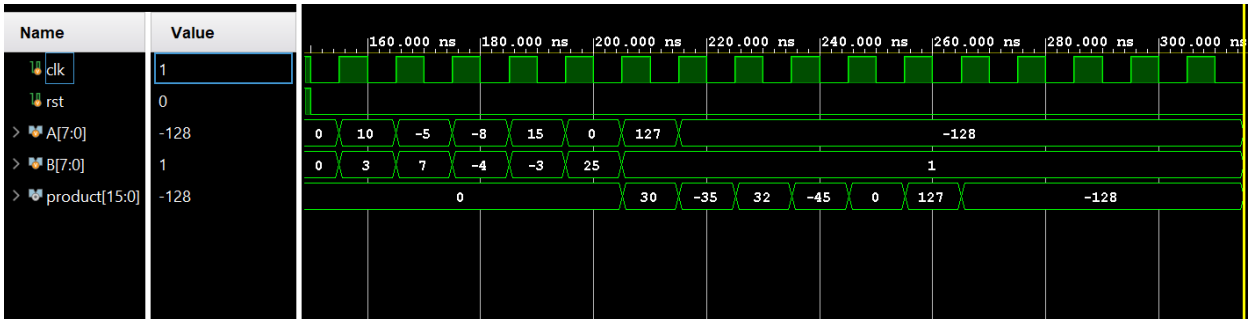


Figure 4: Post-Synthesis Functional

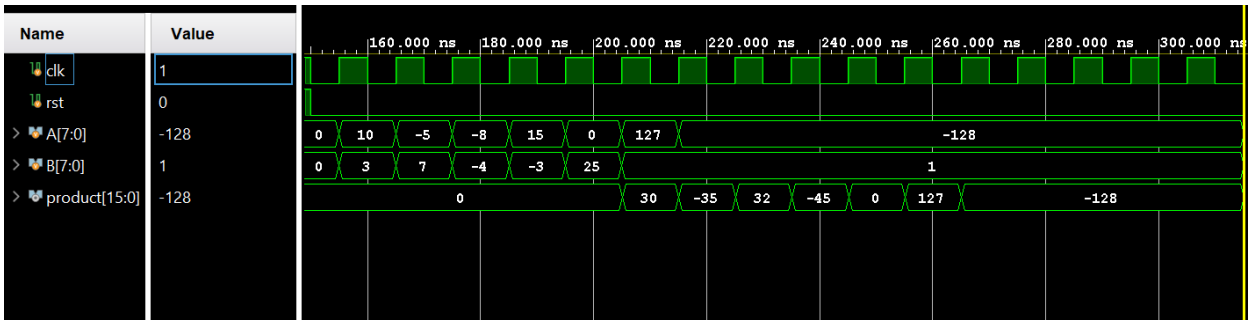


Figure 5: Post-Synthesis Timing

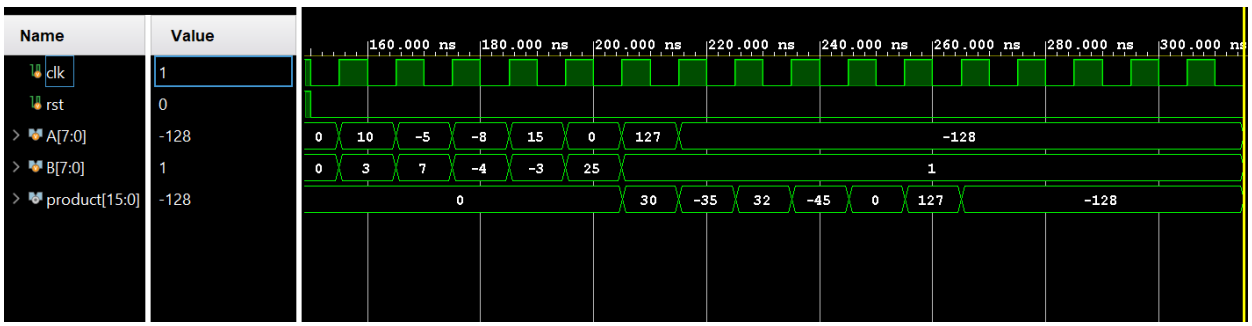


Figure 6: Post-Implementation Functional

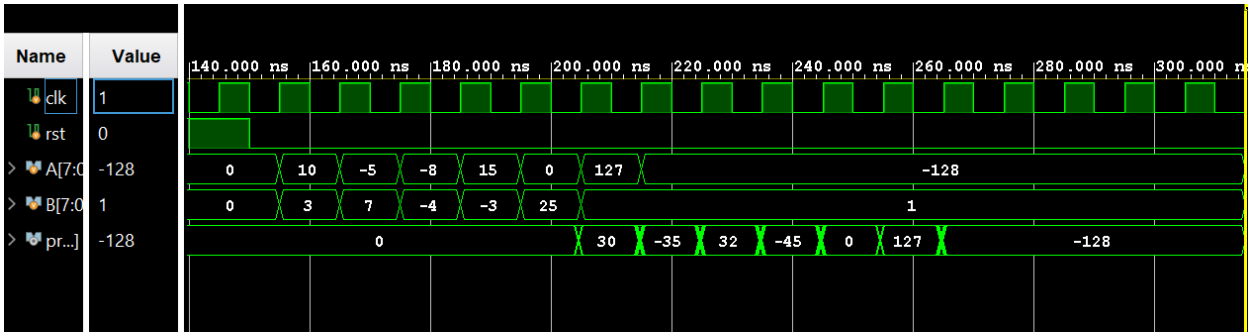


Figure 7: Post-Implementation Timing

2.3 Performance

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	6.929 ns	Worst Hold Slack (WHS):	0.131 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	120	Total Number of Endpoints:	120	Total Number of Endpoints:	137

All user specified timing constraints are met.

Figure 8: Timing Summary (Pipelined)

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	3.422 ns	Worst Hold Slack (WHS):	0.346 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	16	Total Number of Endpoints:	16	Total Number of Endpoints:	33

All user specified timing constraints are met.

Figure 9: Timing Summary (Unpipelined Radix-4 Booth Multiplier)

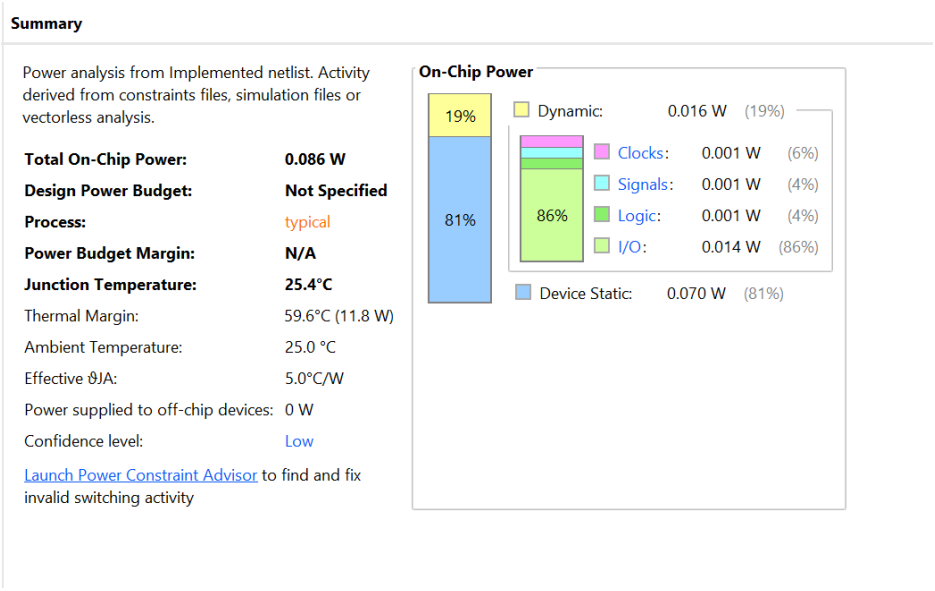
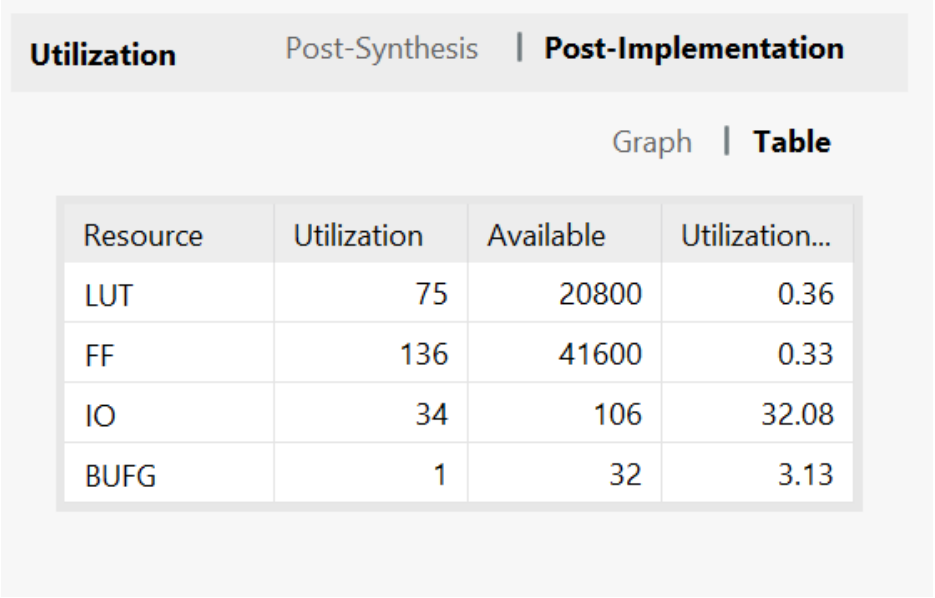


Figure 10: Power Report



Resource	Utilization	Available	Utilization...
LUT	75	20800	0.36
FF	136	41600	0.33
IO	34	106	32.08
BUFG	1	32	3.13

Figure 11: Resource Utilization (Pipelined)

Utilization Post-Synthesis | **Post-Implementation**

Graph | **Table**

Resource	Utilization	Available	Utilization %
LUT	94	20800	0.45
FF	32	41600	0.08
IO	34	106	32.08
BUFG	1	32	3.13

Figure 12: Resource Utilization (Unpipelined Radix-4 Booth Multiplier)

Clock: 10 ns **WNS:** 6.299 ns

[$T_{critical} = 10 - 6.299 = 3.701$ ns $\Rightarrow$ $f_{max} \approx 270.2$ MHz]

Max Frequency (Pipelined): 270.2 MHz

Max Frequency (Unpipelined): 152.0 MHz

Latency (Pipelined): 5 clock cycles

Latency (Unpipelined): 2 clock cycles

Comparison with Unpipelined Design

- **Area:** Pipelined design uses slightly more resources due to additional pipeline registers.
- **Delay:** Reduced critical path delay in pipelined architecture.
- **Throughput:** Significantly higher in pipelined design as multiple operations are overlapped.
- **Latency:** Pipelined = 5 cycles, Unpipelined = 2 cycles.

3 Up-Down Counter

3.1 Without Debouncing

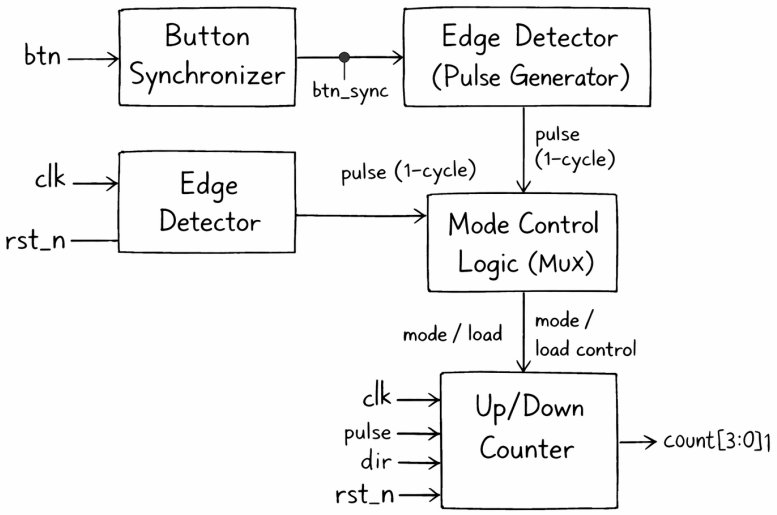


Figure 13: Hand-drawn Design

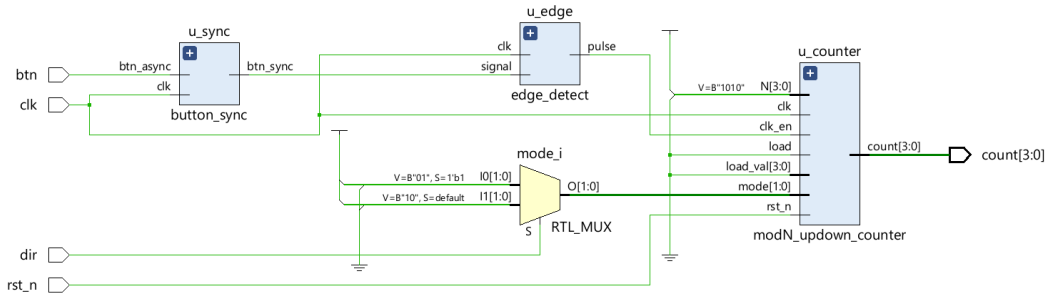


Figure 14: RTL

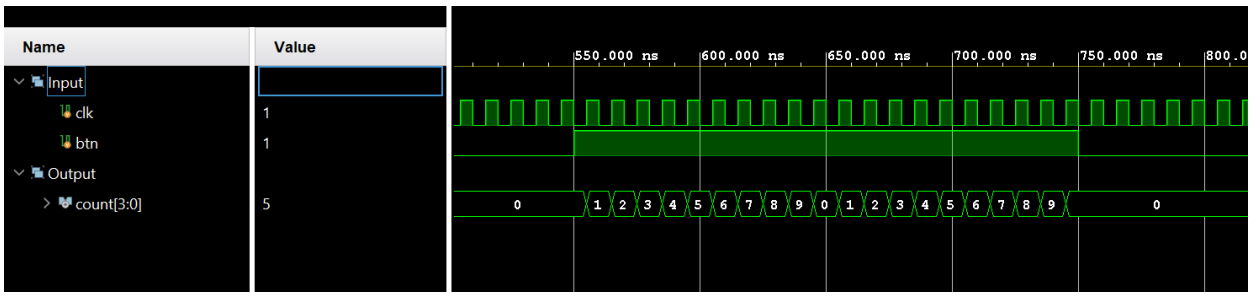


Figure 15: Behavioral

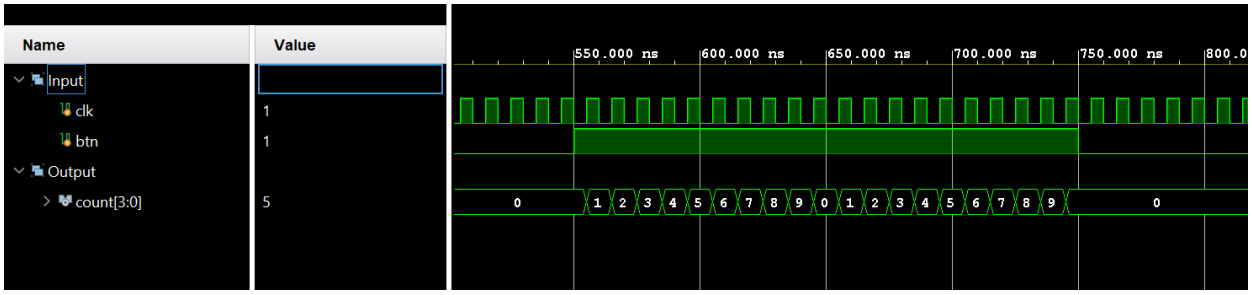


Figure 16: Post-Synthesis Functional

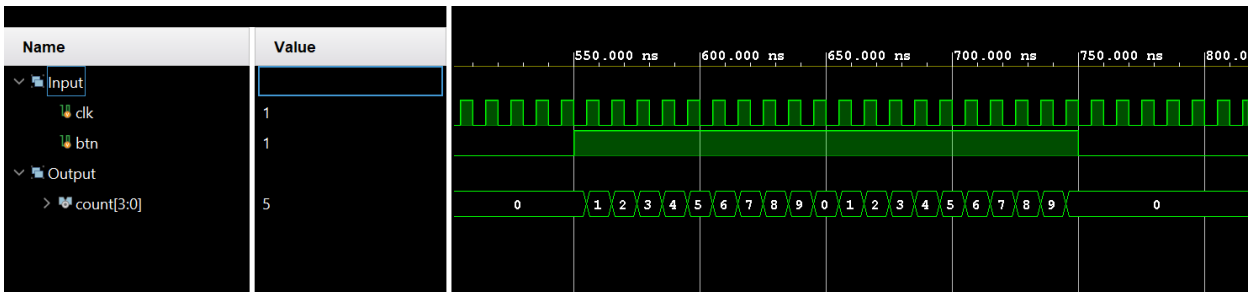


Figure 17: Post-Synthesis Timing

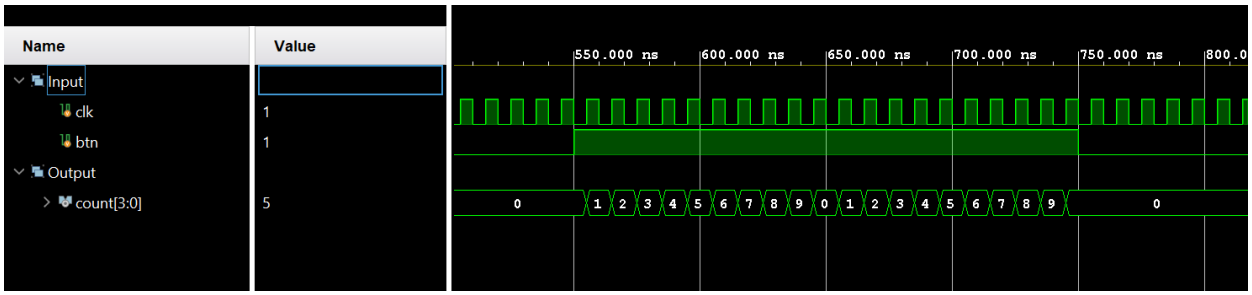


Figure 18: Post-Implementation Functional

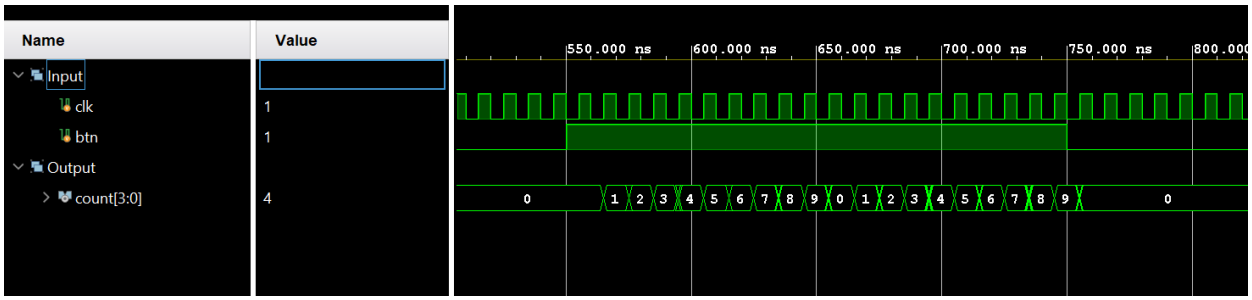


Figure 19: Post-Implementation Timing

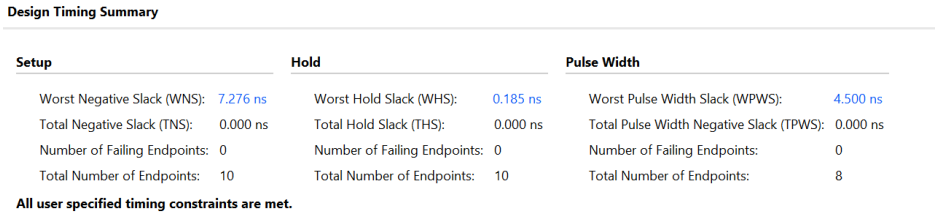


Figure 20: Timing Summary

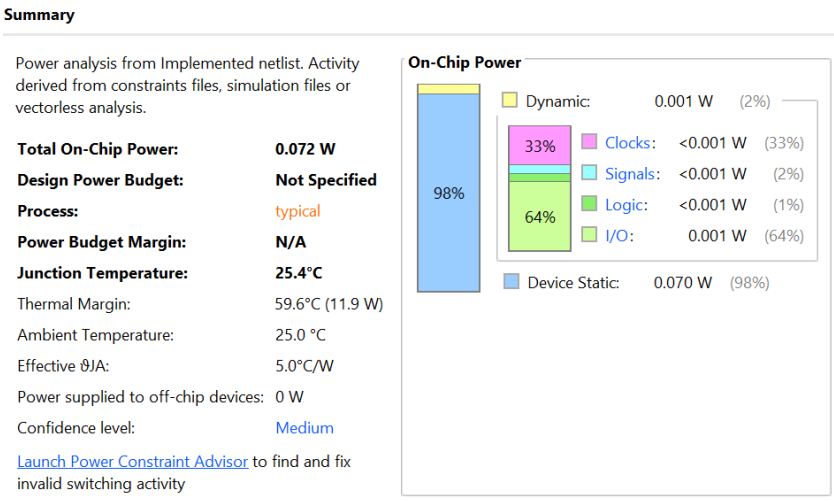


Figure 21: Power

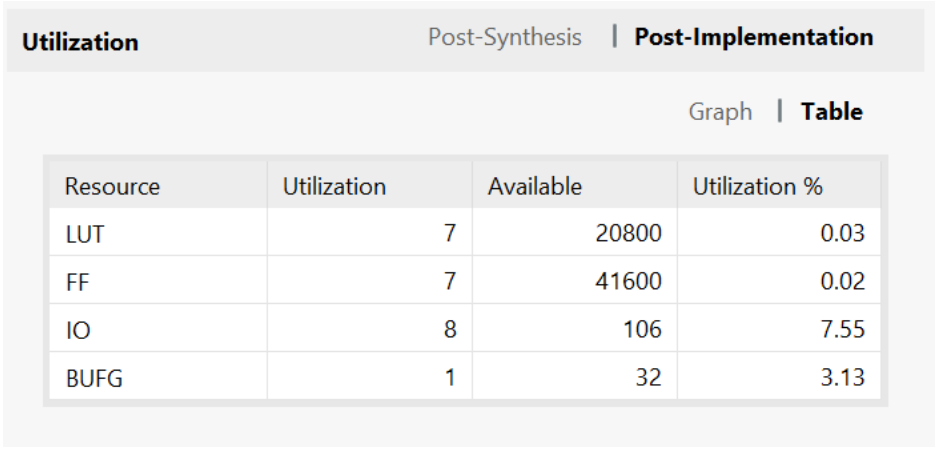


Figure 22: Utilization

WNS: 7.276 ns

[$T_{critical} = 10 - 7.276 = 2.724$ ns $\Rightarrow$ $f_{max} \approx 367.1$ MHz]

Max Frequency: 367.1 MHz

Observation: Multiple counts due to switch bouncing and asynchronous input effects.

3.2 With Debouncing

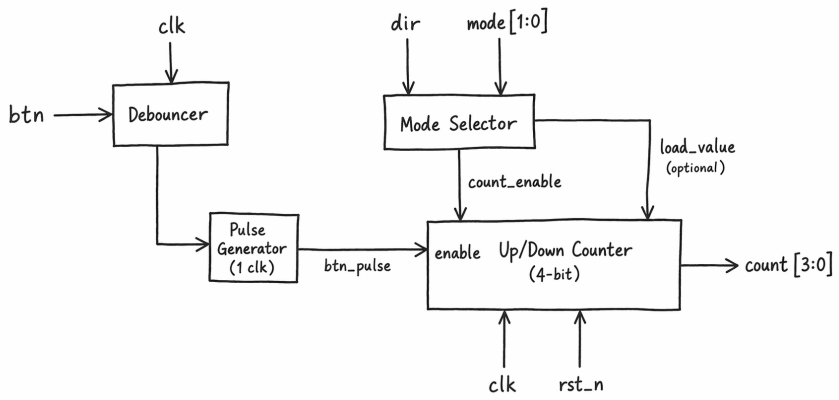


Figure 23: Hand-drawn Design

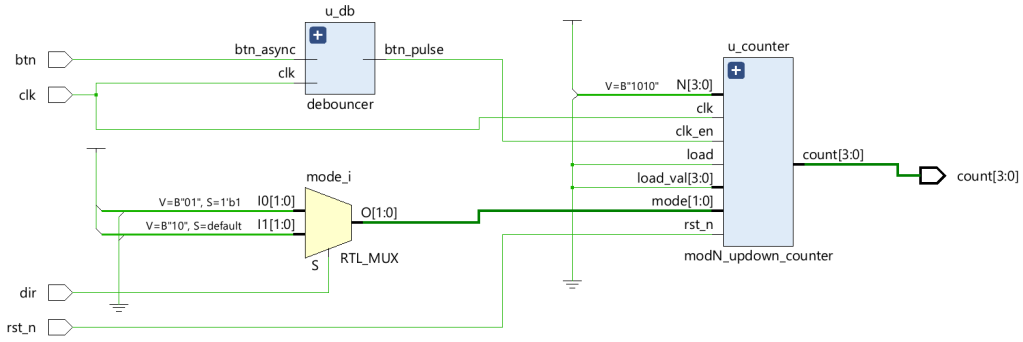


Figure 24: RTL

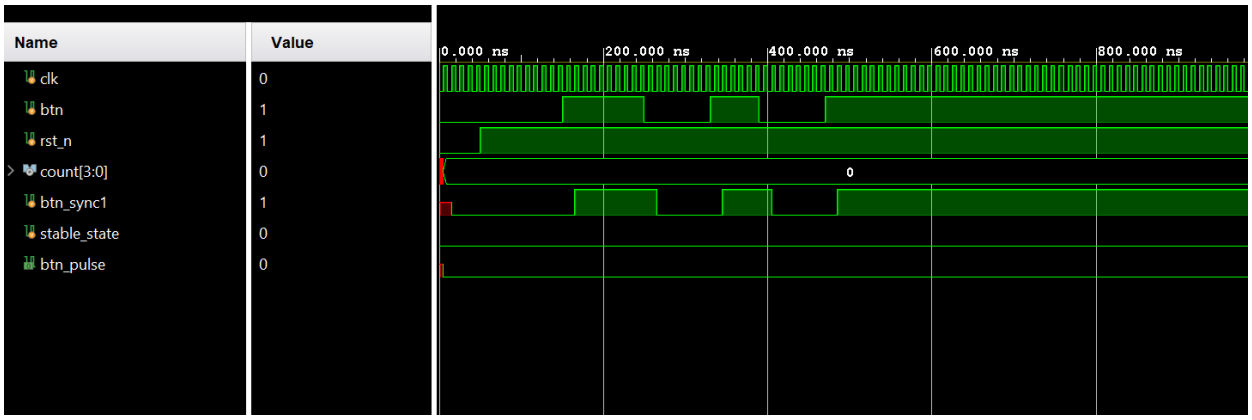


Figure 25: Initial (Count unaffected)

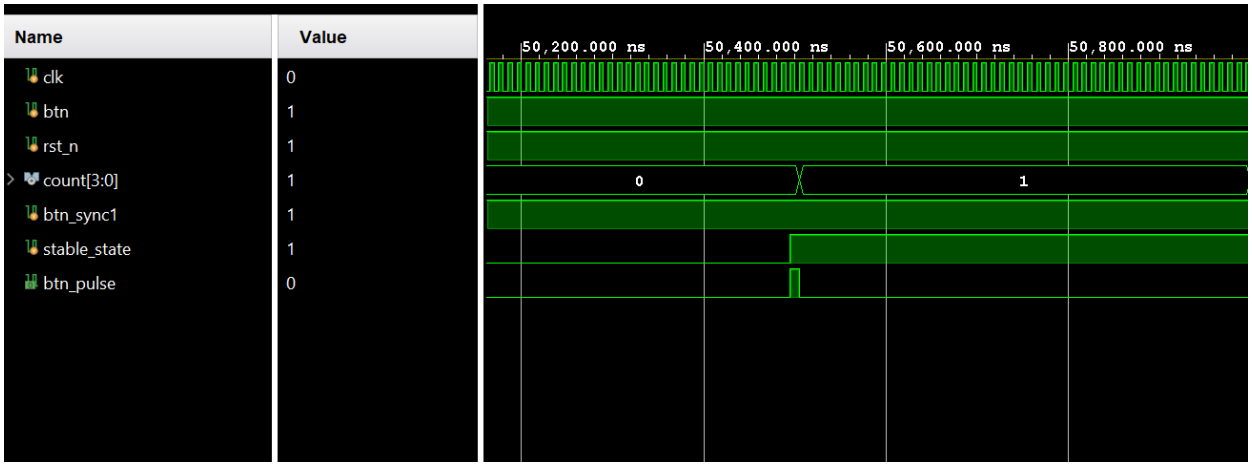


Figure 26: Behavioral

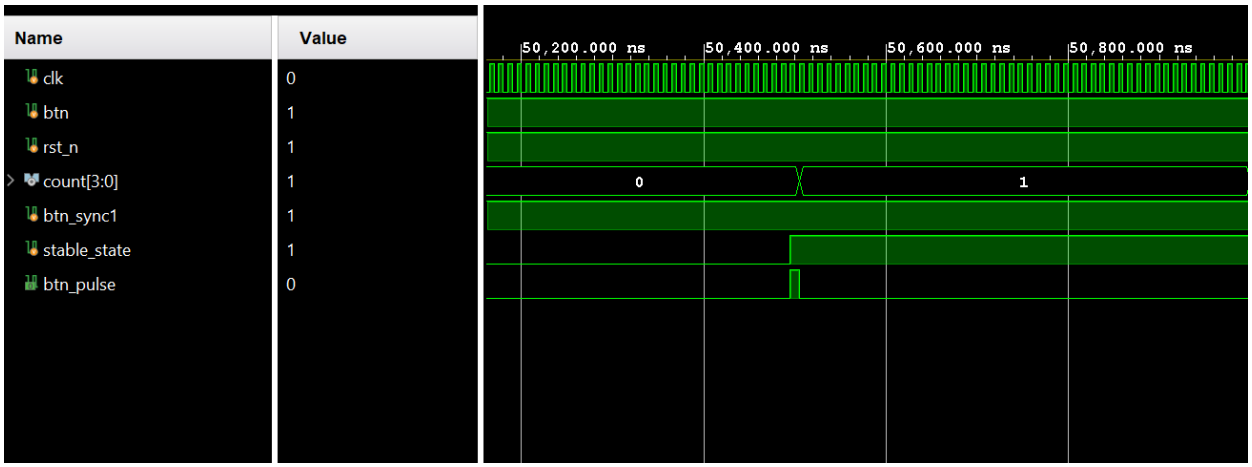


Figure 27: Post-Synthesis Functional

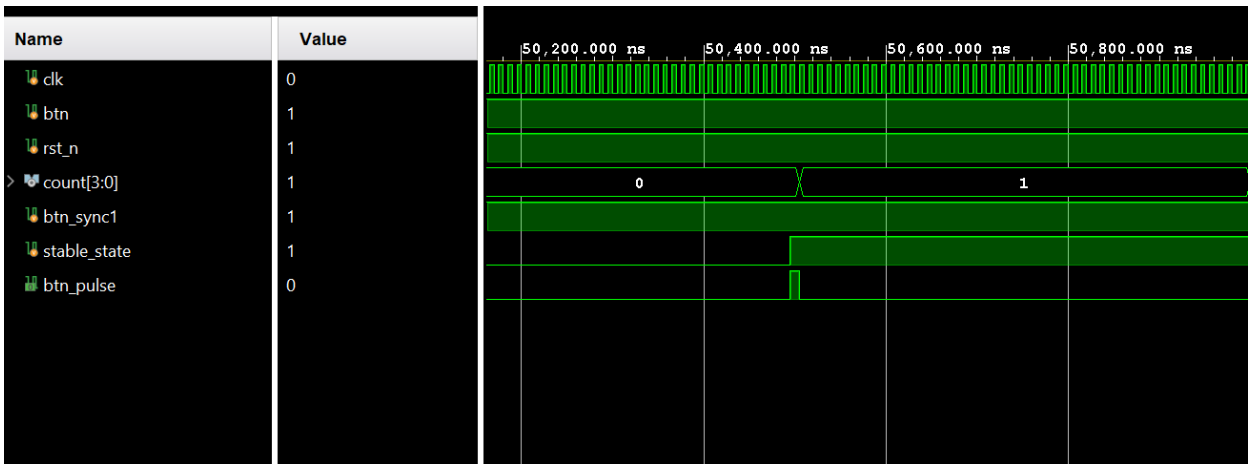


Figure 28: Post-Synthesis Timing

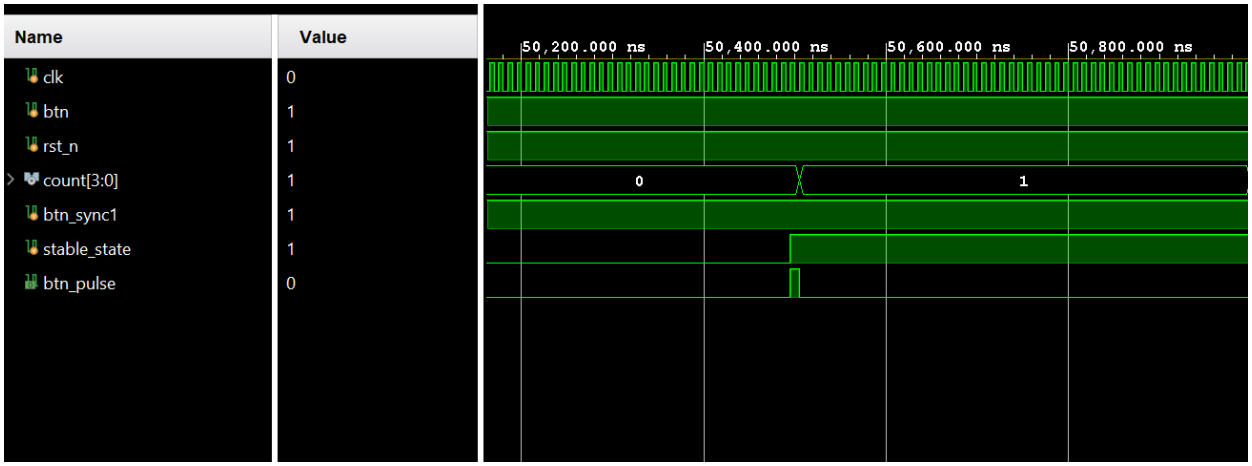


Figure 29: Post-Implementation Functional

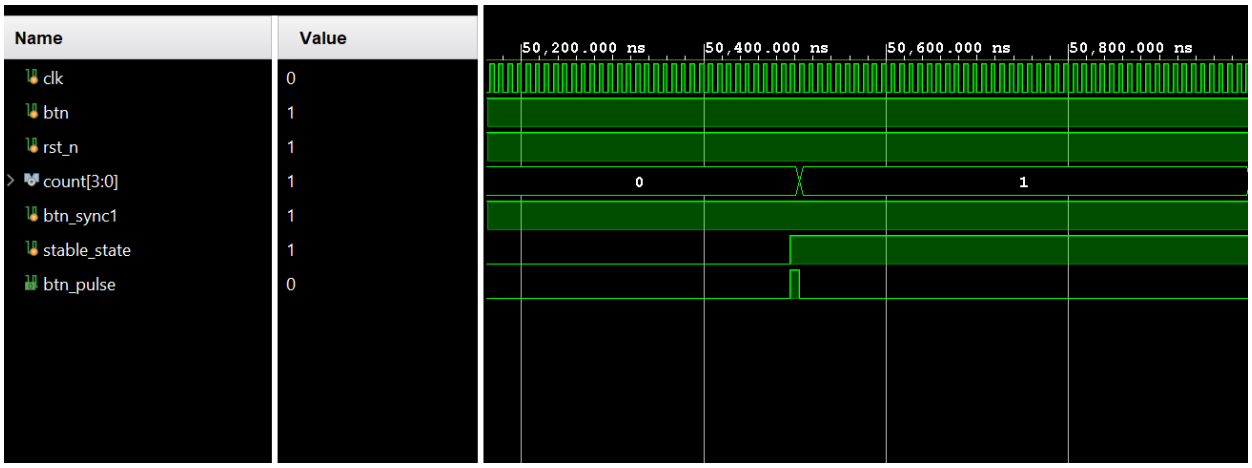


Figure 30: Post-Implementation Timing

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	6.420 ns	Worst Hold Slack (WHS):	0.168 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	37	Total Number of Endpoints:	37	Total Number of Endpoints:	22

All user specified timing constraints are met.

Figure 31: Timing Summary

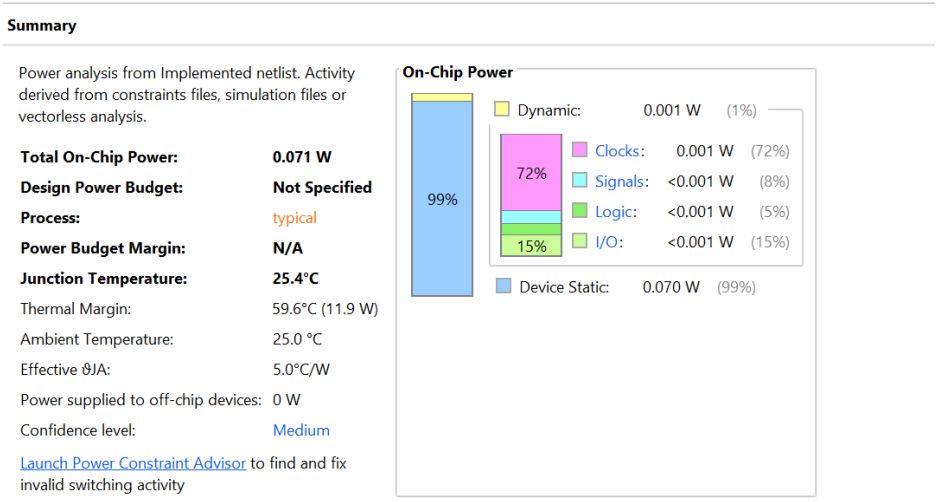
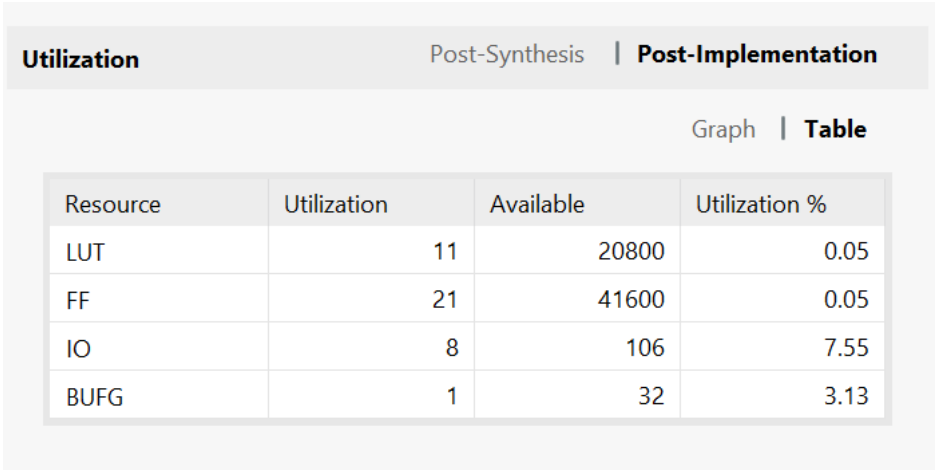


Figure 32: Power



Resource	Utilization	Available	Utilization %
LUT	11	20800	0.05
FF	21	41600	0.05
IO	8	106	7.55
BUFG	1	32	3.13

Figure 33: Utilization

WNS: 6.420 ns

[$T_{critical} = 10 - 6.420 = 3.580$ ns $\Rightarrow$ $f_{max} \approx 279.3$ MHz]

Max Frequency: 279.3 MHz

Observation: Debouncing produces a single clean pulse per press, ensuring reliable counting.

4 Conclusion

This work demonstrated the impact of pipelining and proper handling of asynchronous inputs in FPGA-based digital designs. The pipelined Radix-4 Booth multiplier achieved a significantly higher maximum operating frequency compared to the unpipelined design due to reduced critical path delay, at the cost of increased latency (5 cycles vs. 2 cycles) and slightly higher resource utilization due to additional pipeline registers. The improvement in throughput highlights the effectiveness of pipelining for high-performance arithmetic units.

In the case of asynchronous inputs, the implementation without debouncing resulted in erroneous multiple counts due to switch bouncing and metastability effects. The addition of synchronization and debouncing logic ensured reliable operation by generating a single clean pulse per input event, thereby improving system robustness.

Overall, the study reinforces that pipelining is essential for performance optimization, while synchronization and debouncing are critical for correctness in real-world FPGA systems.

5 References

References

[1] A. D. Booth, "A Signed Binary Multiplication Technique," *Quarterly Journal of Mechanics and Applied Mathematics*, vol. 4, no. 2, pp. 236–240, 1951.

[2] Xilinx Inc., "Vivado Design Suite User Guide: Synthesis (UG901)," 2023.

[3] S. Brown and Z. Vranesic, *Fundamentals of Digital Logic with Verilog Design*, 3rd ed., McGraw-Hill, 2014.

[4] W. J. Dally and J. W. Poulton, *Digital Systems Engineering*, Cambridge University Press, 1998.

[5] Clifford E. Cummings, "Clock Domain Crossing (CDC) Design Verification Techniques Using SystemVerilog," SNUG, 2008.